

Element B student work

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<https://www.fisheries.noaa.gov/protecting-marine-life>

The NOAA Fisheries "Protecting Marine Life" page summarizes laws, programs, and science—like the Endangered Species Act, species recovery planning, permits and authorizations, research surveys, population assessments, and marine life-in-distress/stranding response—that directly apply to sea turtles and their conservation. That information is useful when designing improvements for turtles in captivity because it highlights legal and ethical requirements (permits, take restrictions, viewing and handling guidelines), points to best-practice recovery and rehabilitation protocols used for stranded or injured turtles, and provides scientific data on health, population trends, and habitat needs that should inform enclosure design, enrichment, diet, veterinary care, and release-readiness. Knowing the threats addressed by NOAA (e.g., bycatch, habitat loss, noise) helps designers create environments and training programs that reduce stress and encourage natural behaviors, while following recovery-plan goals ensures captive programs support long-term conservation outcomes.

<https://polarjournal.net/marine-sanctuaries-hope-for-whales-dolphins-in-captivity/#:~:text=SEA%20LIFE%20TRUST%20has%20been,up%20to%20eight%20additional%20animals>

PolarJournal's "Marine Sanctuaries – Hope for Dolphins and Whales" describes the current three sanctuaries in Canada, Iceland, and the USA. The article highlights key issues of these animals' well-being when dolphins and whales are used as entertainment devices, which has caused entertainment shows with these animals to be almost non-existent these days. All three programs offered a more naturalistic approach to their caring of these animals as they provided real seawater to the animals and allowed the whales and dolphins to interact with other animals in their habitat.

<https://awionline.org/content/confinement-marine-life>

The Animal Welfare Institute article highlights several general welfare principles. The articles mentioned that the animals must be able to perform natural behaviors (roaming, foraging, basking, social interactions), enclosure size and complexity should match species' ecological needs rather than human convenience, and captive programs should prioritize rehabilitation and release or truly sanctuary-like care rather than entertainment or breeding for display.

The article's emphasis on how confinement, removal from social groups, and forced interactions can cause stress and abnormal behaviors suggests designers should provide species-specific environmental enrichment, opportunities for natural foraging (including live or simulated prey), thermal gradients and basking areas for terrestrial/semiaquatic species, and sufficient space and substrate for digging or nesting. AWI's cautious stance—supporting

captivity only in exceptional, welfare-focused circumstances and exploring seaside sanctuaries—also argues for independent welfare monitoring, limits on performance-driven training, and planning for long-term physical and psychological needs

<https://ballardbrief.byu.edu/issue-briefs/mistreatment-of-wild-animals-in-captivity>

The Ballard Brief article outlines the broad causes, consequences, and remedies for mistreatment of wild animals in captivity—points that map directly onto designing better turtle enclosures and care. It highlights how captivity itself, cramped or unnatural enclosures, poor water quality, inappropriate diets, forced interactions with visitors, inadequate social arrangements, and weak oversight or staff training all produce chronic stress, stereotypic behaviors, health problems, and premature death—warnings you can use to avoid common pitfalls when designing for turtles. The piece also summarizes practical mitigation strategies (meaningful environmental enrichment, species-appropriate habitat features, careful breeding policies, strong monitoring and veterinary care, and sanctuary-style housing or carefully managed reintroductions) and notes the limits of release programs and regulatory gaps, which are important when planning long-term welfare and compliance. For turtles specifically, that translates into giving far more water volume and depth than minimums, natural substrate and basking/UV facilities, varied foraging-based feeding and enrichment.

<https://aqua.org/stories/2025-07-15-seal-rescue-recap>

The National Aquarium's "Seal Rescue Recap" summarizes a busy rescue season, highlighting how the Stranding Response Center and the Animal Care and Rescue Center handled many young, medically fragile marine mammals through triage, long-term rehabilitation, targeted medical treatments (antibiotics, wound care, tube feedings), hands-on foraging lessons, and post-release monitoring with satellite tags, while relying on partner organizations and limited funding and permits. Although the article focuses on seals, those operational details are directly useful for designing improvements for turtles in captivity: we could plan for seasonal influxes and spare capacity, build a dedicated triage and quarantine workflow to prevent disease spread, and incorporate age-specific hands-on care and nutrition support.

